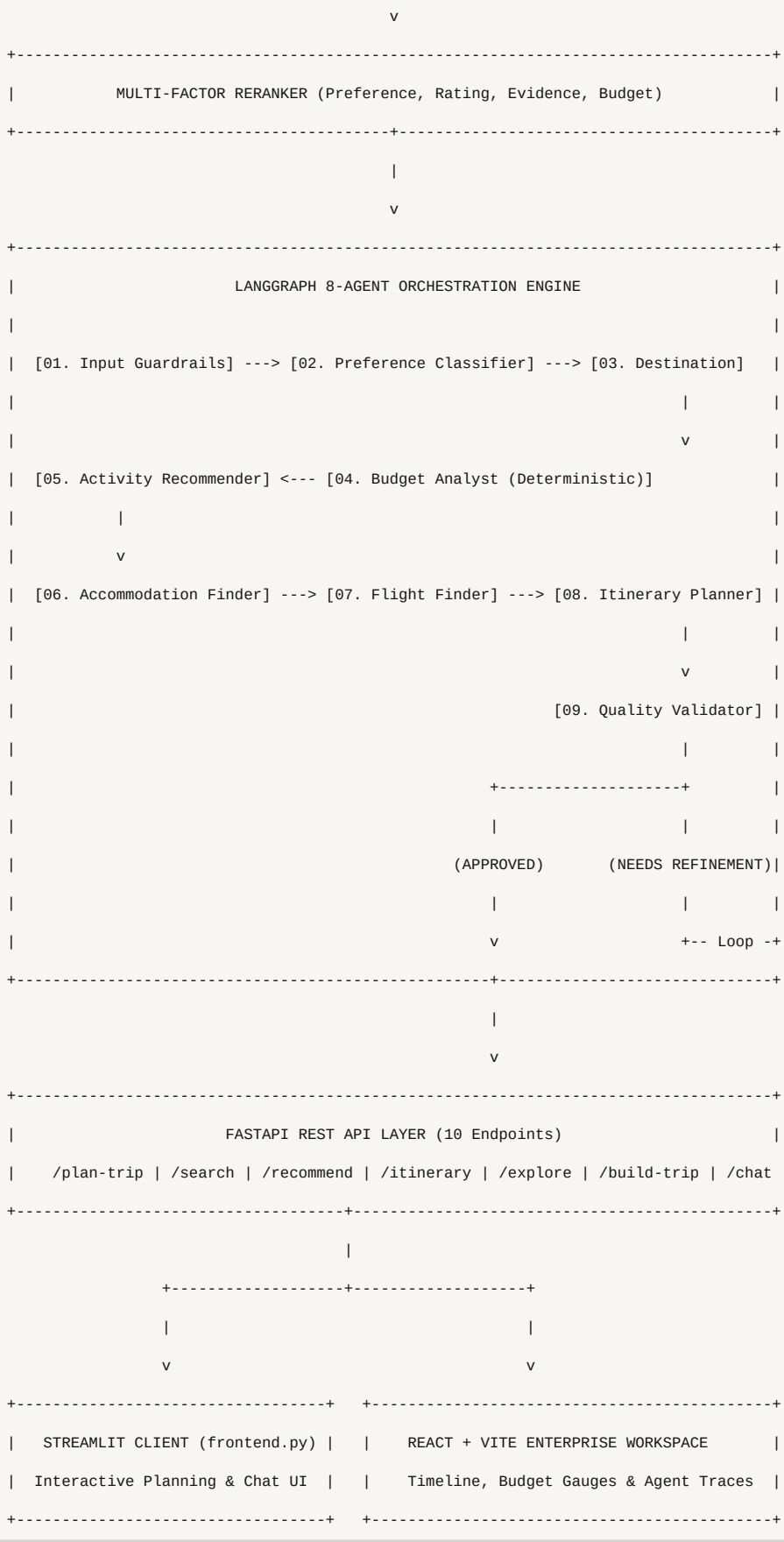
System Architecture & Data Flow Diagram

AI-Powered Travel Planning & Itinerary Assistant (Waypoint)

1. End-to-End System Data Flow

The diagram below illustrates the end-to-end multi-agent retrieval and reasoning pipeline. Raw UCI Dataset 485 records (5,456 reviews across 24 categories) are ingested, preprocessed into structured documents, embedded, and indexed into ChromaDB. At query time, hybrid retrieval fuses dense semantic vectors with BM25 lexical keyword ranks via Reciprocal Rank Fusion (RRF), passed to the 8-Agent LangGraph engine with live OpenStreetMap POI grounding.


MULTI-FACTOR RERANKER (Preference, Rating, Evidence, Budget)
LANGGRAPH 8-AGENT ORCHESTRATION ENGINE
[01. Input Guardrails] ---> [02. Preference Classifier] ---> [03. Destination]
[05. Activity Recommender] <--- [04. Budget Analyst (Deterministic)]
[06. Accommodation Finder] ---> [07. Flight Finder] ---> [08. Itinerary Planner]
[09. Quality Validator]
(APPROVED)
(NEEDS REFINEMENT)
Loop
FASTAPI REST API LAYER (10 Endpoints)
/plan-trip | /search | /recommend | /itinerary | /explore | /build-trip | /chat
STREAMLIT CLIENT (frontend.py)
Interactive Planning & Chat UI
REACT + VITE ENTERPRISE WORKSPACE
Timeline, Budget Gauges & Agent Traces

2. Layer Responsibilities

Component Layer	Technology	Key Responsibility
Data Storage & RAG	ChromaDB + BM25	Indexes 5,456 UCI reviews; supports hybrid rank fusion retrieval.
Embeddings	all-MiniLM-L6-v2	384-dimensional dense semantic vectors with fast cosine similarity.
Knowledge & Grounding	OpenStreetMap (Photon/Nominatim)	Dynamic live lookup for real tourist attractions and hotels worldwide.
Multi-Agent Flow	LangGraph + StateGraph	Orchestrates 8 typed agents with autonomous self-refinement feedback loop.
LLM Reasoning	Google Gemini / AI Credits	Interprets intent, curates schedules, and evaluates plan quality as LLM Judge.
API Gateway	FastAPI + Uvicorn	Exposes 10 validated async REST endpoints with OpenAPI 3.1 documentation.
Streamlit Interface	Streamlit (frontend.py)	Interactive single-command UI for explore, plan, and travel chat.
React Workspace	React 18 + Vite + TypeScript	Editorial planning workspace with day timeline, gauges, and agent traces.